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Physics 2.0

Formalisation and Physlib: a synergy between AI, mathematics and physics

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github.com/leanprover-community/physlib josephtoobysmith.com

section 1 of 6

The problem we face

The problem
Truth
Trust
Community
Recognition
Conclusion

The problem we face

1. AI is getting better and better.
2. We don’t know how far it will get.
3. The arXiv and journals are being overwhelmed.
4. It is hard to know what to trust and what not to.

Evidence that AI is getting better

july 2026

The Jacobian conjecture

Keller, 1939. Counterexample in $\mathbb{C}^3$.
Alpoge, with Claude Fable 5

august 2026

Ten open problems

From maths and CS
OpenAI “Astra”

september 2026

Navier–Stokes problem

Finite time blow-up.
AI models

october 2026

372 problems

OpenAI, internal model

Things to preserve: four tenets of academia

Truth

Trust

Community

Recognition

Physics 1.0

truth

Disseminated in academic papers.

trust

Peer review.

community

Spends its time writing papers, and reviewing them.

recognition

Citation counts, and numbers of papers.

What I think

"Optimism is a duty. The future is open."

— Karl Popper

1. We should optimistically push towards a solution.

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1. We should optimistically push towards a solution.
2. Here I will present my solution, an idea I call **Physics 2.0**.

What I think

"Optimism is a duty. The future is open."
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1. We should optimistically push towards a solution.
2. Here I will present my solution, an idea I call **Physics 2.0**.
3. When doing so, we should bear in mind what AI now makes **possible**.

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Reorganising truth

The problem

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What is wrong with papers?

1. Hard to find information.
2. Hard to reuse.
3. Hard to find connections.
4. Papers are outdated. It is time to find something new.

The solution

Store information not in papers, but in an **open-source, community-run** corpus of knowledge.

Easy to **find** information.

Easy to **reuse**.

Easy to find **connections**.

Similar reason to why **SciPy** exists in Python.

Dissent against our solution

1. Such a corpus of knowledge is hard to maintain.
2. It is hard to keep up to date.

AI helps with **all of this**.

Ingredient 1 of Physics 2.0



leanprover-community / Physlib

A community-driven project to digitalise results from physics into Lean.

public

lean 4

★ **786** stars **205** forks **114** contributors started 2024

I will explain the details later.

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Adjusting trust

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What is wrong with peer review?

1. It does not scale efficiently with AI.
2. Mistakes get into papers, and slip past reviewers.

The mathematicians have a solution

It is called **formalisation**.

Technical interlude

What is an interactive theorem prover?

A programming language in which you can write mathematical theorems proofs, and definitions, with mathematical **correctness** and **consistency** guaranteed. And the definitions you write are **executable**.

What is formalisation?

The process of writing results into an **ITP**.

The four main interactive theorem provers

Rocq **Isabelle/HOL** **Lean** **Agda**

This talk is in **Lean**.

A rapid overview of Lean

Proof compiler

Definitions, theorems, proofs
⟶ **consistency & correctness, guaranteed**

Built on type theory

Lean knows the **type** of everything, and checks that things have the type expected.

$$\underbrace{p}_{\text{proof}} : \underbrace{P}_{\text{theorem}} \qquad \underbrace{7}_{\text{term}} : \underbrace{\mathbb{N}}_{\text{type}}$$

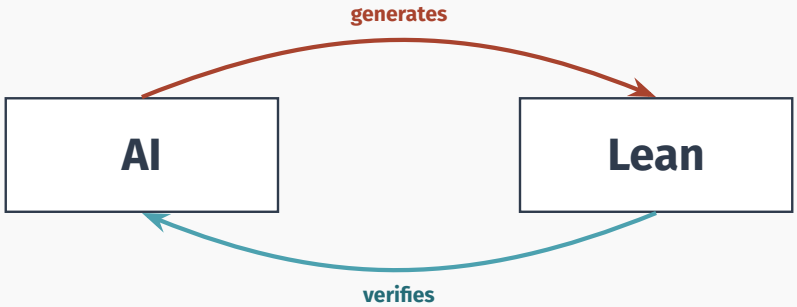
An example

```
def energy (m v : Real) : Real :=
  m * v ^ 2 / 2

theorem energy_nonneg
    (m v : Real) (hm : 0 <= m) :
    0 <= energy m v := by
  unfold energy
  positivity
```

Open this example in the live editor

The formalisation–AI loop



End of technical interlude

How good is AI-driven formalisation now?

It can essentially formalise **anything you want**.

Anthropic's Claude formalised Fermat's Last Theorem in Lean in about eleven days, on Lean's three standard axioms alone and with no `sorry`s.

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However, this does not necessarily mean **well**.

Formalisation does not give you complete trust

Formalisation is not the whole picture, and should not give you the whole trust.

Fermat's Last Theorem, made trivial

```
-- "the integers"
def MyInt : Type := Empty

def add (a b : MyInt) : MyInt :=
  Empty.elim a
def pow (a : MyInt) (n : Nat) : MyInt :=
  Empty.elim a

theorem flt (a b c : MyInt)
    (n : Nat) (hn : 3 <= n) :
    Ne (add (pow a n) (pow b n)) (pow c n) :=
  Empty.elim a
```

Lean accepts it. Nothing here is about numbers.

Ingredient 2a of Physics 2.0

Physlib is a **corpus of knowledge** whose content is **formalised into Lean**.

Community run, and open source.

a dual process

Lean verification

the mathematics

and

Human review

the physics context, style, organisation and quality

The problem

Lean verification takes the mathematics. But part of the trust still rests on **human review** — and that **cannot scale**.

Ingredient 2b of Physics 2.0

There is another solution to scaling trust: **one-look review**, similar to the arXiv.

PhyslibAlpha

This is **PhyslibAlpha**, and it allows things to **scale**.

Project structure

upstream **Mathlib**

core

Physlib

Curated to a high standard for long-term reuse.

rapid

PhyslibAlpha

Lighter review, for large-scale AI-assisted contributions.

qi

QuantumInfo

Quantum information theory, being brought closer to Physlib.

adjacent **CSLib**

Physlib is one of three main ones



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mathlib4

The math library of Lean 4.

★ 4,309 ⑂ 1,768
👥 798 🕒 2021



leanprover-community /

Physlib

Physics, digitalised into Lean. Community-driven.

★ 786 ⑂ 205
👥 114 🕒 2024



leanprover /

CSLib

The computer science library of Lean.

★ 754 ⑂ 214
👥 50 🕒 2025

Evidence that this can be done at scale and works

Top-level directories of `Physlib/`, by lines of Lean.

Relativity	27,545	StatisticalMechanics	2,449
QFT	20,162	ProbabilisticTheory	2,306
Particles	19,104	Cosmology	1,858
SpaceAndTime	16,069	Meta	1,773
Mathematics	15,795	FluidDynamics	1,361
QuantumMechanics	13,057	CondensedMatter	1,003
ClassicalMechanics	11,784	ClassicalFieldTheory	554
Electromagnetism	9,442	Thermodynamics	432
StringTheory	5,285	Optics	48
Units	4,076	LatticeQFT	22

154,125 lines of Lean across **619** files.

Evidence that this can be done at scale and works

The Higgs boson, in Lean

live.lean-lang.org

Additional benefits of having Physlib

1. Allows verification of results, both human and AI.
2. Allows easy reuse, and improvement.
3. Allows assumptions to be tracked.
4. Allows one to easily explore and constrain model spaces.
5. Prevents bit-rot: one corpus, kept compiling.
6. Avoids duplicated work, and definitions that are incompatible with one another.
7. AI speed and efficiency makes this feasible.

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Community

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What the community currently does

Writes papers.
Reviews for journals.

What will they do in this new world?

Writing papers, and reviewing for journals, is exactly the work that AI is now **capable of**.

Ingredient 3 of Physics 2.0

Working on the corpus of knowledge — **Physlib**.

- Review
- Maintenance
- Refactoring
- Foundational work
- Formalisation
- Documentation
- Extending
- Adding experiments
- Building tools

People are already doing this

100+
contributors

7
maintainers

Always looking for more

Built for the community

we refactor

When a better approach is found, we take it.

a choice of starting point

There is no single correct axiomatisation, so we allow a **choice of starting point**.

Difficulties in formalising physics

1. Validation does not come with the guarantees of verification.
2. One physical concept, many mathematical descriptions.
3. Physics papers aren't written as definitions and theorems.
4. How physical assumptions enter.

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Recognition

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The problem with current recognition

Recognition is currently based only on **publishing papers** and **citations**.

In this new world that is a **poor proxy** for the work that actually matters — because it is precisely the work we have seen AI is **very capable of**.

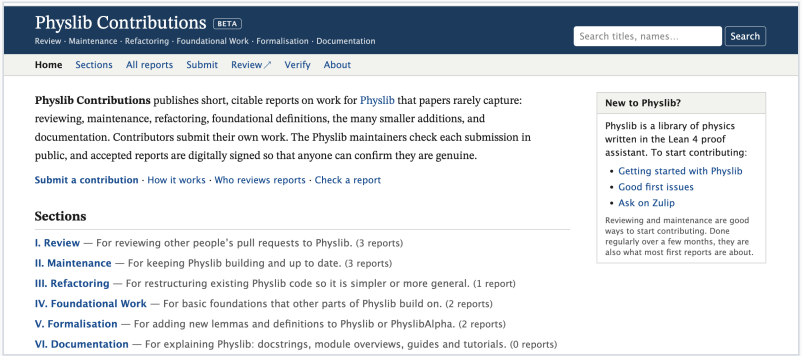
Who gets the recognition?

The library is the **foundation of truth**.

A separate, **arXiv-like system** records contributions — **verified**, and usable on a CV.

Ingredient 4 of Physics 2.0

contributions.physlib.io



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Conclusion

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Physics 2.0

truth

Papers → **Physlib**

trust

Peer review → **Lean + human review**

community

Writing papers → **Building and maintaining the corpus**

recognition

Citations → **Contributions recorded and verified**

Call to action

if you are hiring, or on grant committees

Start to recognise contributions to projects like **Physlib**, **Mathlib** and **CSLib** on a par with papers.

if you are an expert in physics

Come to **Physlib**, help review, help push forward your area of expertise.

"Optimism is a duty. The future is open. It is not predetermined. No one can predict it, except by chance. We all contribute to determining it by what we do. We are all equally responsible for its success."

— Karl Popper, *The Lesson of This Century*

Thank you for watching

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